6

Formal constraints

The difference between informal and formal constraints is one of degree. Envision a continuum from taboos, customs, and traditions at one end to written constitutions at the other. The move, lengthy and uneven, from unwritten traditions and customs to written laws has been unidirectional as we have moved from less to more complex societies and is clearly related to the increasing specialization and division of labor associated with more complex societies.[1]

The increasing complexity of societies would naturally raise the rate of return to the formalization of constraints (which became possible with the development of writing), and technological change tended to lower measurement costs and encourage precise, standardized weights and measures. The creation of formal legal systems to handle more complex disputes entails formal rules; hierarchies that evolve with more complex organization entail formal structures to specify principal/agent relationships. The general characteristics of the shift from status to contract have been amply discussed, but it is worth emphasizing the following.

Formal rules can complement and increase the effectiveness of informal constraints. They may lower information, monitoring, and enforcement costs and hence make informal constraints possible solutions to more

[1]For a lengthy and thoughtful discussion of what we mean by formal rules, see Elinor Ostrom (1986). Ostrom breaks down the rule structure into the following specifics: position rules that specify a set of positions and how many participants hold each position, boundary rules that specify how participants are chosen to hold these positions and how participants leave these positions, scope rules that specify the set of outcomes that may be affected and the external inducements and/or costs assigned to each of these outcomes, authority rules that specify the set of actions assigned to a position at a particular node, aggregation rules that specify the decision functions to be used at a particular node to map action into intermediate or final outcomes, and information rules that authorize channels of communication among participants in positions and specify the language and form in which the communication will take place.

complex exchange (see Milgrom, North, and Weingast, 1990, and Chapter 7 for elaboration). Formal rules also may be enacted to modify, revise, or replace informal constraints. A change in the bargaining strength of parties may lead to an effective demand for a different institutional framework for exchange, but the informal constraints stand in the way of accomplishing it. Sometimes (but not always) it is possible to supersede the existing informal constraints with new formal rules (this point will be elaborated and qualified in Chapter 10).

I

Formal rules include political (and judicial) rules, economic rules, and contracts. The hierarchy of such rules, from constitutions, to statute and common laws, to specific bylaws, and finally to individual contracts defines constraints, from general rules to particular specifications. And typically constitutions are designed to be more costly to alter than statute laws, just as a statute law is more costly to alter than individual contracts. Political rules broadly define the hierarchical structure of the polity, its basic decision structure, and the explicit characteristics of agenda control. Economic rules define property rights, that is the bundle of rights over the use and the income to be derived from property and the ability to alienate an asset or a resource. Contracts contain the provisions specific to a particular agreement in exchange.

Given the initial bargaining strength of the decision-making parties, the function of rules is to facilitate exchange, political or economic. The existing structure of rights (and the character of their enforcement) defines the existing wealth-maximizing opportunities of the players, which can be realized by forming either economic or political exchanges. Exchange involves bargains made within the existing set of institutions, but equally the players at times find it worthwhile to devote resources to altering the more basic structure of the polity to reassign rights.

The extent of economic and political diversity of interests will, given relative bargaining strength, influence the rules' structure. The immediate reason is that the more numerous the interests, the less likely the simple majority (in the polity) will obtain and the more likely exchange will be structured to facilitate complex forms of exchange (partly formal but also partly informal) and other ways of solving problems by coalition formation. It is important to note, however, that the function of formal rules is to promote certain kinds of exchange but not all exchange. Thus Madison, in *Federalist Paper Number 10,* maintained that the constitutional structure was devised in 1787 not only to facilitate certain kinds of exchange, but also to raise the costs of those kinds of exchange that promote the interests of factions. Similarly, in economic exchange patent laws and

trade secret laws are designed to raise the costs of those kinds of exchange deemed to inhibit innovation.

Before going further, it is important to stress that there is nothing in my argument so far about rules that implies efficiency. As stressed above, rules are, at least in good part, devised in the interests of private well-being rather than social well-being. Hence, rules that deny franchise, restrict entry, or prevent factor mobility are everywhere evident. This is not to deny that ideas and norms matter, but to establish that as a first approximation, rules are derived from self-interest.

Rules are generally devised with compliance costs in mind, which means that methods must be devised to ascertain that a rule has been violated, to measure the extent of the violation (and consequent damages to the party to exchange), and to apprehend the violator. The costs of compliance include measuring the multiple attributes of the goods or services being exchanged and measuring the performance of agents. In many cases, the costs of measurement, given the technology of the time, exceed the gains, and rules are not worth devising and ownership rights are not delineated. Changes in technology or relative prices will alter the relative gains from devising rules.

With these generalizations as background, we can now use the framework derived from Chapters 3 and 4 to describe more closely political rules, property rights (economic rules), and contracts.

II

Broadly speaking, political rules in place lead to economic rules, though the causality runs both ways. That is, property rights and hence individual contracts are specified and enforced by political decision-making, but the structure of economic interests will also influence the political structure. In equilibrium, a given structure of property rights (and their enforcement) will be consistent with a particular set of political rules (and their enforcement). Changes in one will induce changes in the other. But because of the priority of political rules, we will analyze the structure of the political system first.

We start with a simplified model of a polity made up of a ruler and constituents.[2] In such a simple setting, the ruler acts like a discriminating monopolist, offering to different groups of constituents protection and justice or at least the reduction of internal disorder and the protection of property rights in return for tax revenue. Because different constituent groups have different opportunity costs and bargaining power with the

[2]This simple model is developed in much more detail in Chapter 3, "A Neo-classical Theory of the State" in North (1981).

ruler, different bargains result. But there are also economies of scale in the provision of these (semipublic) goods of law and enforcement. Hence, total revenue is increased, but the division of incremental gains between ruler and constituents depends on their relative bargaining power; changes at the margin, either the violence potential of the ruler or the opportunity costs of the constituent, will result in redivisions of the incremental revenue. Moreover, the ruler's gross and net revenue differ significantly as a result of the necessity of developing agents (a bureaucracy) to monitor, meter, and collect the revenue. All the consequences inherent to agency theory obtain here.

This model of the polity becomes one step more complicated when we introduce the concept of a representative body reflecting the interests of constituent groups and their role in bargaining with the ruler. This concept, consistent with the origin of parliaments, estates general, and cortes in early modern Europe, reflects the needs of the ruler to get more revenue in exchange for which he or she agrees to provide certain services to constituent groups. The representative body facilitates exchange between the parties. On the ruler's side, this leads to the development of a hierarchical structure of agents, which is a major transformation from the simple (if extensive) management of the king's household and estates to a bureaucracy monitoring the wealth and/or income of the king's constituents.

When we move from the historical character of representation in early modern Europe to modern representative democracy, our story is complicated by the development of multiple interest groups and by a much more complicated institutional structure devised to facilitate (again given relative bargaining strength) the exchange between interest groups.[3] This political transaction cost analysis is built on the recognition of the multiplicity of interest groups reflecting concentrations of voters in particular locations. Thus in the United States polity, there are elderly in Florida and Arizona, miners in Pennsylvania and West Virginia, artichoke growers in California, automobile manufacturers in Michigan, and so forth. Because

[3]The development of political theory in the last twenty-five years has paralleled developments in economic theory. The developments began in an a-institutional setting, in which the model paralleled the a-institutional model of economics. But the result, in terms of the formal theory, was that no stable equilibrium would evolve and that *cycling* would be a continuous pattern of political systems (at least in two-party, nonideological models). However, this formal finding was at odds with empirical and descriptive studies that provided no evidence of such disequilibrating characteristics, and it remained to take a further step in political theory to explore the nature of the institutional structure that provided for the evolution of equilibrium states in the political system. For a description of this evolution and a model of structure-induced equilibrium, see Kenneth Shepsle, "Institutional Equilibrium and Equilibrium Institutions" (1986).

there are multiple interest groups, no particular interest group that a legislator may represent can form a majority. Therefore, legislators cannot succeed acting alone, but must make agreements with other legislators, with different interests.

What kinds of institutions will evolve from exchange relationships between legislators reflecting multiple interest groups? Previous work, beginning with Buchanan and Tullock (1962), focused on vote-trading or logrolling. This approach was a step forward in recognizing the way by which legislators can strike bargains that facilitate exchange; however, it is too simple to solve fundamental problems involved in legislative exchange. It assumes that all bills and payoffs were known in advance, and it has a timeless dimension to it. In fact, a variety of exchanges arise in which today's legislation can only be enacted by commitments made for a future date. To lower the costs of exchange, it was necessary to devise a set of institutional arrangements that would allow for exchange over space and time. As with the economic exchange described in Chapter 4, the problem is to measure and enforce the exchange of rights.

How does credible commitment evolve to enable agreements to be reached when the payoffs are in the future and on completely different issues? Self-enforcement is important in such exchange, and in repeat dealings a reputation is a valuable asset. But as in economic exchange, the costs of measurement and enforcement, discovering who is cheating whom, when free-riding will occur, and who should bear the cost of punishing defectors make self-enforcement ineffective in many situations. Hence political institutions constitute ex ante agreements about cooperation among politicians. They reduce uncertainty by creating a stable structure of exchange. The result is a complicated system of committee structure, consisting of both formal rules and informal methods of organization. The evolution of this structure in the U.S. Congress is described in a recent study of the structure by Barry Weingast and William Marshall entitled "The Industrial Organization of Congress" (1988). In their conclusion, Weingast and Marshall specify the kind of structure that evolved:

> Instead of trading votes, legislators exchanged special rights affording the holder of these rights additional influence over well-defined policy jurisdictions. This influence stems from the property rights established over the agenda mechanisms, that is, the means by which alternatives arise for votes. The extra influence over particular policies institutionalizes a specific pattern of trades. When the holders of seats on committees are precisely those individuals who would bid for votes on these issues in a market for votes, policy choice under the committee system parallels that under a more explicit exchange system. Because the exchange is institutionalized, it need not be renegotiated each new legislative session, and it is subject to fewer enforcement problems. (Weingast and Marshall, 1988, p. 157)

The evolution of polities from single absolute rulers to democratic governments is typically conceived as a move toward greater political efficiency. In the sense that democratic government gives a greater and greater percentage of the populace access to the political decision-making process, eliminates the capricious capacity of a ruler to confiscate wealth, and develops third-party enforcement of contracts with an independent judiciary, the result is indeed a move toward greater political efficiency. But it would be wrong to assert that the result is efficient political markets in the same sense as we mean efficient economic markets. The existence of efficient economic markets entails competition so strong that, via arbitrage and information feedback, one approximates the Coase zero transaction cost conditions. Such markets are scarce enough in the economic world and even scarcer in the political world. It is true that the move toward a democratic polity will reduce legislative transaction costs per exchange (as elaborated by Weingast and Marshall, 1988), but not only will the number of exchanges increase so that the size of the total political transaction sector will grow, the agency costs between constituent and legislator and legislator and bureaucrat will be substantial. Moreover, *rational ignorance* on the part of constituents is going to increase the role, in many situations, of incomplete subjective perceptions playing an important part in choices. The atypical informed constituent may indeed know his or her own interest in making choices about familiar *local* repeated problems, but even the informed constituent is going to be at sea in making choices about the complex nonrepetitive problems of an interdependent political and economic world. The point is that formal political rules, like formal economic rules, are designed to facilitate exchange but democracy in the polity is not to be equated with competitive markets in the economy. The distinction is important with respect to the efficiency of property rights.

III

As a first approximation we can say that property rights will be developed over resources and assets as a simple cost-benefit calculus of the costs of devising and enforcing such rights, as compared to the alternatives under the status quo. Changes in relative prices or relative scarcities of any kind lead to the creation of property rights when it becomes worthwhile to incur the costs of devising such rights. This simple model has been the basis not only for my own early work (North and Thomas, 1973) but also for a substantial amount of the property rights literature, which looks on the development of property rights as a simple function of changes in economic costs and benefits. The simple model of the evolution of property rights would be consistent with Axelrod's *The Evolution of Coopera-*

tion (1984), but such an argument leaves out the role of the polity and the consequent kinds of property rights that will be specified and enforced.

In North (1981), I revised the 1973 argument to account for the obvious persistence of inefficient property rights. These inefficiencies existed because rulers would not antagonize powerful constituents by enacting efficient rules that were opposed to their interests or because the costs of monitoring, metering, and collecting taxes might very well lead to a situation in which less efficient property rights yielded more tax revenue than efficient property rights. This argument is an improvement over the efficiency argument but needs amplification.

The efficiency of the political market is the key to this issue. If political transaction costs are low and the political actors have accurate models to guide them, then efficient property rights will result. But the high transaction costs of political markets and subjective perceptions of the actors more often have resulted in property rights that do not induce economic growth, and the consequent organizations may have no incentive to create more productive economic rules. At issue is not only the incremental character of institutional change, but also the problem of devising institutions that can provide credible commitment so that more efficient bargains can be struck. In Chapter 11, I shall explore how such sufficient paths of development can persist through time.

IV

The rules descend from polities to property rights to individual contracts. Contracts will reflect the incentive-disincentive structure imbedded in the property rights structure (and the enforcement characteristics); thus the opportunity set of the players and the forms of organization they devise in specific contracts will be derived from the property rights structure.

The contract specified by economic theory is simple, complete, and straightforward. It involves an exchange of a unidimensional product at an instant of time. The contract in modern complex economies both is multidimensional and extends over time. Because there are multiple dimensions, with respect both to the physical characteristics and to the property rights characteristics of the exchange, of necessity the result is that one must spell out many of the provisions. Moreover, the contract will typically be incomplete, in the sense that there are so many unknowns over the life of contracts extending over time that the parties will (deliberately) leave to the courts or to some third party the settlement of disputes that arise over the life of the contract.[4]

[4]See Goldberg (1976) for discussion of relational exchange and the complicated contracts that in fact characterize modern exchange.

Contracts provide not only an explicit framework within which to derive empirical evidence about the forms of organization (and hence are the basic empirical source for testing hypotheses about organization), but also clues with respect to the way by which the parties to an exchange will structure more complex forms of organization. That is, the contracts will reflect different ways to facilitate exchange, whether through firms, franchising, or other more complex forms of agreement that extend in a continuum from straightforward market exchange to vertically integrated exchange.[5] The complex interaction of institutional constraints and the development of organizations are the subject of Chapter 9.

I should close this chapter with a word of warning – although explicit rules provide us with a basic source of empirical materials by which to test the performance of economies under varying conditions, the degree to which these rules have unique relationships to performance is limited. That is, a mixture of informal norms, rules, and enforcement characteristics together defines the choice set and results in outcomes. Looking only at the formal rules themselves, therefore, gives us an inadequate and frequently misleading notion about the relationship between formal constraints and performance.

[5]The enormous literature that has evolved in the last fifteen or twenty years in the New Industrial Organization has provided us with an immense amount of valuable material on the kinds of organizations that will evolve and the forms of governance that will be reflected as ways to solve problems of complicated exchange. See in particular Oliver Williamson, *Markets and Hierarchies* (1975), and the subsequent literature that has developed from Williamson's pioneering work.

7

Enforcement

A good deal of literature on transaction costs takes enforcement as a given, assuming either that it is perfect or that it is *constantly* imperfect. In fact, enforcement is seldom either, and the structure of enforcement mechanisms and the frequency and severity of imperfection play a major role in the costs of transacting and in the forms that contracts take. There are two reasons why enforcement is typically imperfect. The first takes us back to the preceding chapters that explore the costs of measuring the multiple margins that constitute contract performance. The second rests in the fact that enforcement is undertaken by agents whose own utility functions influence outcomes.

In Chapter 4, asymmetries of information held by principals and agents about the valuable attributes of what was being exchanged were examined in the context of the wealth-maximizing behavior of the parties to exchange. In this chapter I wish to extend that analysis to explore the problems that arise in the transfer of rights. Parties to an exchange must be able to enforce compliance at a (transaction) cost such that the exchange is worthwhile to them. On the face of it, the problem sounds simple. Surely the gains from trade, which economists take to be the bedrock of economic performance, should make it worthwhile to evolve cooperative solutions among parties to capture jointly those gains. Indeed under certain circumstances, as I have noted in earlier chapters, the issues are so resolved. Trade does exist, even in *stateless* societies. Yet, as emphasized earlier, the inability of societies to develop effective, low-cost enforcement of contracts is the most important source of both historical stagnation and contemporary underdevelopment in the Third World.[1]

[1]This emphasis upon enforcement is another major difference between Oliver Williamson's approach to transaction costs and the one taken in this study. Williamson assumes enforcement to be imperfect (otherwise opportunism would never pay), but does not make it an explicit variable in his analysis. Such an approach simply does not lead the scholar to be able to deal with the problems of historical evolution, where

I

Under what conditions will contracts tend to be self-enforcing? In a wealth-maximizing world, the answer can be stated very simply. Contracts will be self-enforcing when it pays the parties to live up to them – that is, in terms of the costliness of measuring and enforcing agreements, the benefits of living up to contracts will exceed the costs. The most likely and indeed empirically observable state in which contracts are self-enforcing is that in which the parties to exchange have a great deal of knowledge about each other and are involved in repeat dealings, as detailed in the earlier chapter dealing with tribal and primitive societies and with small communities. Under these conditions, it simply pays to live up to agreements. In such a world, the measured costs of transacting are very low because of a dense social network of interaction. Cheating, shirking, opportunism, all problems of modern industrial organization, are limited or indeed absent because they do not pay. Norms of behavior determine exchange and formal contracting does not exist.

At the other extreme, the world of impersonal exchange is characterized by specialized interdependence in which the well-being of individuals depends upon the complex structure characterized by individual specialization and hence exchange extends through both time and space. In a pure model of the world of impersonal exchange, goods and services or the performance of agents is characterized by many valued attributes, exchange takes place over time, and there are no repeat dealings. In the context of a wealth-maximizing world, where there are high costs of measurement and no form of enforcement is possible, the gains from cheating and reneging exceed the gains from cooperative behavior. I have stated, of course, an extreme form of impersonal exchange, because in the real world, whether present or past (where impersonal exchange did occur to a degree), we find all kinds of mitigating circumstances by which parties attempt to assure compliance. The exchange of hostages, ostracism of merchants who reneged on agreements, to name two examples, provide incentives to parties to live up to agreements. Reputations, depending on the costs of information, provided parties in long-distance trade and impersonal exchange a mechanism to enforce agreements. Kinship ties, various forms of loyalty, minority groups in societies bound together by common beliefs in a hostile world – all provided frameworks within which living up to agreements was worthwhile. In addition, it should be noted that at times and places ideological commitments to integrity and honesty also played a major role. Nevertheless, the dilemma

the key problems of institutional change, of contracting, and of performance turn on the degree to which contracts can be enforced between parties at low cost.

that is posed by impersonal exchange without effective third-party enforcement is central to the major issues of development.

II

Let me explain this dilemma more precisely by expanding on the game theory framework briefly outlined in Chapter 2.[2] There I started with a very simple prisoner's dilemma situation in which it simply did not pay parties in a single exchange or in a single agreement to live up to the terms of exchange. Under these conditions, rational self-interested individuals will arrive at a Pareto inferior solution, that is, one that leaves both parties worse off than they would have been had they cooperated, because it is determined by one of the parties that he or she will be much worse off if the other party chooses not to cooperate. Now, there are ways for the parties to get around such a dilemma. They might ex ante sign a contract agreeing to live up to some set of standards, and they might hire lawyers to see that the contract is enforced. Note that by so doing, however, they have introduced transaction costs to the argument. And the costs of transacting, that is of providing the mechanism for assuring that the parties live up to a cooperative solution, may exceed the gains that the parties can derive from it.

If we shift from a once and for all game to a repeated game or an iterated game, then the possibility of a cooperative solution becomes much more evident, as Axelrod has explored (1984). That is, if the game continues indefinitely, it usually pays the parties to live up to the terms of exchange, because the gains from successive iterations exceed the benefits that could be derived from a single defection, from "running off with the profits." Note, however, that a game so conceived must be played in perpetuity. If there is an end of the game or people believe that the game might end, then indeed the discount rate may enter in to determining whether it is worthwhile to continue to cooperate. The smaller the probability of continuing for another round, the greater must be the payoffs to sustain an equilibrium; also, the greater the possibility of short-run gains, the greater must be the payoffs. Note that if the game runs continuously, there are still transaction costs, because one must still acquire information about the other party. However, the assumptions that are involved in this cooperative solution are seldom realized in the real world. They entail the game lasting continuously, they entail that one repeats the game with the

[2]This section depends heavily upon a substantial literature in game theory of which the essay "Corporate Culture and Economic Theory" by David M. Kreps (forthcoming in Alt and Shepsle, editors, *Perspectives on Positive Political Economy*) was especially useful.

same players, and they entail that one can observe compliance on the part of the other party. Observing compliance in terms of the model suggests that one can measure unambiguously the outcomes of contracts, so that one can determine whether a party has in fact not lived up to the terms of the contract.

This evolving story of game theory tells us that under very simplified conditions, that is, when the parties acquire perfect information and the game both lasts indefinitely into the future and is played between the same parties, one can reach self-enforcing cooperative solutions. But needless to say, these assumptions not only are strong but are simply not observed in the real world. In a world of impersonal exchange, we are exchanging with multiple individuals and can acquire very little information about all of them. Our information not only is imperfect, but varies remarkably from one party to another. Many times the exchange is a once and for all exchange and not repeated at all. Under these conditions, it is easy to see why the problems that we have laid out here are simply unresolvable in terms of cooperative solutions that can exist in impersonal exchange. The inevitable conclusion that one arrives at in a wealth-maximizing world is that complex contracting that would allow one to capture the gains from trade in a world of impersonal exchange must be accompanied by some kind of third-party enforcement. Indeed, this conclusion clearly mirrors the quotation from Norman Schofield at the end of Chapter 2, describing the conditions necessary for equilibrium solutions to emerge in the context of complex, cooperative games.

This game theoretic story can be translated into the framework developed in the preceding chapters. In the straightforward neoclassical story, the gains from trade are realized with zero transaction costs. That is, the parties to exchange costlessly know everything about the other party and enforcement is perfect. No institutions are necessary in a world of complete information. With incomplete information, however, cooperative solutions will break down unless institutions are created that provide sufficient information for individuals to police deviations. There are two parts to an institution's assuring cooperation. First, it is necessary to form a communications mechanism that provides the information necessary to know when punishment is required. By making available the relevant information, institutions make possible the policing of defections. Typically they economize on information, so, for example, players need no longer know the entire past history of any partner. Second, because punishment is often a public good in which the community benefits but the costs are borne by a small set of individuals, institutions must also provide incentives for those individuals to carry out punishment when called on to do so (see Milgrom, North, and Weingast, 1990, for an elaboration of this argument). It should be stressed that creating an in-

stitutional environment that induces credible commitment entails the complex institutional framework of formal rules, informal constraints, and enforcement that together make possible low-cost transacting.

This argument states that the players may devise an institutional framework to improve measurement and enforcement and therefore make possible exchange, but the resultant transaction costs raise the costs of exchange above the neoclassical level. The more resources that must be devoted to transacting to assure cooperative outcomes, the more diluted are the gains from trade of the neoclassical model. The more complex the exchange in time and space, the more complex and costly are the institutions necessary to realize cooperative outcomes. Quite complex exchange can be realized by creating third-party enforcement via voluntary institutions that lower information costs about the other party; ultimately, however, viable impersonal exchange that would realize the gains from trade inherent in the technologies of modern interdependent economies requires institutions that can enforce agreements by the threat of coercion. The transaction costs of a purely voluntary system of third-party enforcement in such an environment would be prohibitive. In contrast there are immense scale economies in policing and enforcing agreements by a polity that acts as a third party and uses coercion to enforce agreements. But therein lies the fundamental dilemma of economic development. If we cannot do without the state, we cannot do with it either. How does one get the state to behave like an impartial third party?

III

If formal third-party enforcement is essential, it is important to define exactly what one means by it. In principle, third-party enforcement would involve a neutral party with the ability, costlessly, to be able to measure the attributes of a contract and, costlessly, to enforce agreements such that the offending party always had to compensate the injured party to a degree that made it costly to violate the contract. These are strong conditions that obviously are seldom, if ever, met in the real world. It is costly to measure the attributes. The enforcer is an agent and has his or her own utility function, which will dictate his or her perceptions about the issues and therefore will be affected by his or her own interests. Enforcement is costly. Indeed, it is frequently costly even to find out that a contract has been violated, more costly to be able to measure the violation, and still more costly to be able to apprehend and impose penalties on the violator.

But achieving third-party enforcement in contracting is a major dilemma for economies that would engage in impersonal exchange. The devel-

opment of credible commitment on the part of political bodies, such that one has assurances that political bodies will not violate contracts of parties or engage in conditions that will alter radically the wealth and income of parties, is always relative; even in the most highly developed countries we observe political entities altering the wealth of parties by all manner of changes in the price level or in rules that affect the well-being of individuals. Nevertheless, there is an immense difference in the degree to which we can rely upon contract enforcement between developed countries and Third World countries.

In developed countries, effective judicial systems include well-specified bodies of law and agents such as lawyers, arbitrators, and mediators, and one has some confidence that the merits of a case rather than private payoffs will influence outcomes. In contrast, enforcement in Third World economies is uncertain not only because of ambiguity of legal doctrine (a measurement cost), but because of uncertainty with respect to behavior of the agent.

Even when enforcement is sufficient to enable elaborate contracts to be made because they are subject to a system of courts that act as a bulwark against the violation of contracts, contractors must take into account those margins at which it is hard to measure whether a contract has been fulfilled. Hence, under conditions of uncertainties with respect to the future or problems of agency for which enforcement is difficult, the contracting parties will attempt to structure contracts that will minimize the potential both for contract violation and for rent dissipation by the parties.[3]

I conclude this analysis of enforcement by pointing out where it is taking us. Third-party enforcement means the development of the state as a coercive force able to monitor property rights and enforce contracts effectively, but no one at this stage in our knowledge knows how to create such an entity. Indeed, with a strictly wealth-maximizing behavioral assumption it is hard even to create such a model abstractly. Put simply, if the state has coercive force, then those who run the state will use that force in their own interest at the expense of the rest of the society. Madison laid out a solution to this problem in *The Federalist Papers* and Vincent Ostrom has expanded on it in his theory of the compound republic (1971); the correct constitutional forms will restrain the tyrannical exercise of political power. William Riker's skepticism, however, still seems appropriate:

> But another main theme is a celebration of the efficacy of constitutional forms in restraining the tyrannical exercise of political power. This is the theme I wish to

[3]See Barzel (1982) for an elaboration of these issues.

address. I guess I want to defend the view he attributes to Woodrow Wilson. My initial tendency, however, as a constitutional theorist raised in the same tradition as Professor Ostrom has always been to agree with his proposition. But everytime I convince myself that I have found an instance in which constitutional forms do make a difference for liberty, my discovery comes apart in my hand. It is, of course, all a matter of the direction of causality. Professor Ostrom believes that at least part of the reason we are a free people is that we have certain constitutional forms; but it may just as easily be the case that the reason we have these constitutional forms is that we are a free people. The question is: Does constitutional structure cause a political condition and a state of public opinion or does the political condition and a state of public opinion cause the constitutional structure? This sounds at first like the chicken and egg problem in which there is no causal direction; but I think that usually there is a cause and that constitutional forms are typically derivative. It seems probable to me that public opinion usually causes constitutional structure, and seldom, if ever, the other way around. As Rousseau contended, it is in the end the law that is written in the hearts of the people that counts. (Riker, 1976, p. 13)

In a subsequent historical chapter, I briefly describe the way in which such constitutional forms evolved in England in the seventeenth century. But although that story describes a successful outcome, it does not give a definitive answer to the question of how it was achieved. It was surely a mixture of formal and informal constraints. Both respect for the law and the honesty and integrity of judges are an important part of this success story. They are self-enforcing standards of conduct, and I believe that they are important. How does one create such self-enforcing constraints? Part of the answer is that creating a system of effective enforcement and of moral constraints on behavior is a long, slow process that requires time to develop if it is to evolve – a condition markedly absent in the rapid transformation of Africa from tribal societies to market economies. The quote from William Riker goes to the heart of the issue of creating effective institutional constraints.

8

Institutions and transaction and transformation costs

It takes resources to define and protect property rights and to enforce agreements. Institutions together with the technology employed determine those transaction costs. It takes resources to transform inputs of land, labor, and capital into the output of goods and services and that transformation is a function not only of the technology employed, but of institutions as well. Therefore, institutions play a key role in the costs of production.

In previous chapters I specified why it is costly to transact and examined the variety of forms that institutional constraints take in constraining human interaction. Still ahead (in Chapter 9), I shall explore the way by which learning and organizations can modify and alter the relationship between institutions and transaction costs (and transformation costs as well). But first, I simply wish to pull together the threads of the argument so far.

A hierarchy of rules – constitutional, statute law, common law (and even bylaws) – together will define the formal structure of rights in a specific exchange. Moreover, a contract will be written with enforcement characteristics of exchange in mind. Because of the costliness of measurement, most contracts will be incomplete; hence informal constraints will play a major roles in the actual agreement. These will include reputation, broadly accepted standards of conduct (effective to the extent that the conduct of the other parties is readily observable), and conventions that emerge from repetitive interactions. The relationship between rights and constraints in an exchange can be illustrated at three levels: first at the level of a single straightforward exchange, second in the more complex relationship involved in the production process, and finally for the economy as a whole.

I

For a single straightforward exchange, I use the transfer of a residential property in the modern United States. This transfer involves a bundle of rights over a physical asset in exchange for a sum of money. The rights are

both legal rights defining what one can do with the property and rights over the physical attributes of the property. The sum of money is a command over resources. Institutions determine how costly it is to make the exchange. The costs consist of the resources necessary to measure both the legal and physical attributes being exchanged, the costs of policing and enforcing the agreement, and an uncertainty discount reflecting the degree of imperfection in the measurement and enforcement of the terms of the exchange. The size of the uncertainty discount will be influenced by such specific-to-the-contract factors as asymmetric information about the condition of the house (known to the seller) and the financial condition of the buyer (known to the buyer), by such communitywide factors as the effectiveness of crime prevention, and by such nationwide factors as the stability of the price level.

In the seller's utility are the price, terms, and security of the contractual obligation, that is, the likelihood that the buyer will live up to the contract ex post. The value of the residence to the buyer is a function not only of price and credit terms but also of the attributes that are transferred with the sale. Some, such as the legal rights that are transferred and the dimensions of the property and house, are easily measured, and some, such as the general features of the property, are readily observed on inspection. But others, such as the maintenance and upkeep costs and the characteristics of neighbors, may be far more difficult to ascertain. Equally, the security of property against default, expropriation, uncertain title, and theft will vary according to the difficulty of ascertaining their likelihood and, therefore, their importance.

Now in the traditional neoclassical paradigm, with perfect information (i.e., zero transaction costs), the value of the asset that is transferred assumes not only perfect information but perfectly secure property rights as well. In that case, because both buyer and seller have been able to ascertain costlessly the value of all the attributes (both physical and property rights) and there is no uncertainty or insecurity of property rights, the standard supply and demand models of housing with zero transaction costs would define the value of the asset. In fact, because all of the above-mentioned attributes influence the value of the residence to the buyer and seller, the smaller the discount from the idealized neoclassical model, the more perfect the market. Institutions in the aggregate define and determine the size of the discount, and the transaction costs that the buyer and seller incur reflect the institutional framework.

The transaction costs of the transfer are partly market costs – such as legal fees, realtor fees, title insurance, and credit rating searches – and partly the costs of time each party must devote to gathering information, to searching, and so forth. Obtaining information about crime rates, police protection, and security systems entails search costs to the buyer.

To the degree that the buyer's utility function is adversely affected by noisy neighbors or pets, it will pay to invest in ascertaining neighborhood characteristics and the norms and conventions that shape neighborhood interactions.

The particular institutions matrix of this housing market consists first of all of a hierarchy of legal rules derived from provisions of the U.S. constitution and the powers delegated to the states. State laws defining the conveyance characteristics of real property, zoning laws restricting which rights can be transferred, common and statute law undergirding, defining, or restricting a host of voluntary organizations – all of these influence transaction costs. Realtors, title insurance, credit bureaus, and savings and loan associations that affect the mortgage market all will be influenced. The efficiency of these organizations is a function of the structure of property rights and enforcement and of the capital market (including voluntary as well as governmental guarantees and subsidies and other instruments that exist in the capital market). Equally important are informal constraints that broadly supplement and reinforce the formal rules. They range from conventions of neighborhood conduct to ethical norms defining degrees of honesty in information exchange between the parties involved.

My description has emphasized institutions that lower transaction costs, but some – such as rules that restrict entry, require useless inspections, raise information costs, or make property rights less secure – in fact raise transaction costs. The above-mentioned economic rules are made in the polity and reflect the bargaining strength of contractors, trade unions, and others in the political market. Because that market is *imperfect*, institutions everywhere are a mixed bag composed of those that lower costs and those that raise them. The U.S. residential housing market is a relatively efficient market in which on balance the institutions induce low-cost transacting.

Returning to the model in Chapter 4, the discount from the frictionless exchange envisioned in economic theory will be greater to the degree that the institutional structure allows third parties to influence the value of attributes that are in the utility function of the buyer. These could include the behavior of neighbors, the likelihood of theft, and the possibility of changes by local authorities in zoning ordinances that may affect the value of the property. The greater the uncertainty of the buyer, the lower the value of the asset. Likewise, the institutional structure will determine the risks to the seller that the contract will be fulfilled or that the seller will be indemnified in case of default. It is worth emphasizing that the uncertainties described above with respect to the security of rights are a critical distinction between the relatively efficient markets of high income countries today and economies in the past as well as those in the Third World today.

II

Let us turn from examining the relationship between institutions and transactions costs in a straightforward exchange, and examine those involved in the productions of goods and services. The institutional framework will affect both transformation and transaction costs; the latter because of the direct connection between institutions and transaction costs, and the former by influencing the technology employed. All the usual problems of measurement and enforcement described in the previous section obtain: the structure of property rights, the effectiveness of the courts and judicial system, and the complementary development of voluntary organizations and norms. Specifically, the firm's entrepreneur must be able to ascertain the quantity and quality of a firm's inputs and outputs. Because these can be obtained costlessly in the neoclassical model, the contrast between a hypothetical neoclassical firm and a real firm is striking. The former was little more than a production function without any costs of organization, supervision, coordination, monitoring, and metering. A real-life firm must purchase inputs that constantly require measurement and metering if it is to produce outputs of constant quality. Variability in quality will, ceteris paribus, adversely affect demand for its products. When there is variability, consumers (or if this is an intermediate good, producers) must devote resources to ascertaining quality; hence producers who can guarantee constant quality will be favored.[1]

Perfect measurement and enforcement are implicitly assumed in what we call efficient factor and product markets, but their existence entails a complex set of institutions that encourage factor mobility, the acquisition of skills, uninterrupted production, rapid and low-cost transmission of information, and the invention and innovation of new technologies. Realizing all these conditions is a tall order never completely filled because, as with the institutions of exchange described above, the actual institutional framework is in fact usually a mixed bag of institutions that promote such productivity-raising activities and institutions that provide barriers to entry, encourage monopolistic restrictions, and impede the low-cost flow of information.

We have only to contrast the organization of production in a Third World economy with that in an advanced industrial economy to be im-

[1]For a fascinating analysis of the significant resource costs the producer must engage in to assure constant quality, see the detailed description of the production of peas in Susan Sheehan's essay "Peas," in *The New Yorker,* June 17, 1973. The trouble that Green Giant went to in attempting to ascertain size, tenderness, and sweetness in the production of peas involved enormous monitoring and metering resources that began in the field and did not end until the cans went off to the retailer.

pressed by the consequences of poorly defined and/or ineffective property rights. Not only will the institutional framework result in high costs of transacting in the former, but insecure property rights will result in using technologies that employ little fixed capital and do not entail long-term agreements. Firms will typically be small (except those operated or protected by the government). Moreover, such mundane problems as an inability to get spare parts or a two-year wait to get a telephone installed will necessitate a different organization of production than an advanced country requires. A bribe sufficient to get quick delivery through the maze of import controls or get rapid telephone installation may exist; but the resultant shadow transaction costs significantly alter relative prices and consequently the technology employed.

Even with the relatively secure property rights that exist in high-income countries, it is possible and indeed frequently the case that a technical combination that involves costly monitoring may be less efficient than a technique that has lower physical output but less variance in the product or lower costs of monitoring workers. Much of the recent transaction cost literature implies that institutions only determine transaction costs and techniques only determine transformation costs, but three different illustrations arising from the interplay between techniques, institutions, transformation costs, and transaction costs make clear that the relationship among them is more complex.

1. A contention of Marxist writers is that deliberate deskilling of the labor force occurred during the early twentieth century. That is, employers adopted capital-intensive technologies that eliminated highly skilled workers and replaced them with semiskilled or unskilled workers. The logic of this charge is that the bargaining power of skilled workers enabled them to disrupt the production process strategically, which, given the "high speed throughput" (Chandler's term) of modern technology was enormously costly. Employers found it reduced total cost over time to introduce technologies that used less-skilled workers who did not have the bargaining power to disrupt production. In this case, a new production technology was introduced to reduce transaction costs.

2. Unitizing an oil field, that is, creating an organization with the coercive power and monitoring authority to allocate the output of an oil field, raised transaction costs (because of the resources devoted to creating and maintaining an organization and then to monitoring compliance). At the same time, unitizing the field reduced transformation costs (the result of more efficient pumping and recovery) to an extent that more than offset the rise in transaction costs (Libecap and Wiggins, 1985). In this case an institutional change raised transaction costs that were more than compensated by lower transformation costs.

3. Andrea Shepard (1987) describes the deliberate policy of a semicon-

ductor manufacturer who licenses the designs of new chips to competitors, so that customers can be assured that the chip manufacturer will not be able to hold up customers who adopt the new design. By alleviating customers' concerns, it enhances demand for the product. Although this policy lowers transaction costs, it does so at the sacrifice of productive efficiency, because both scale economies and learning curve effects are lost to competing firms. Second sourcing appears to be a common practice.

Informal constraints frequently play a major role with respect to the quantity and quality of labor output. Although Marxists long ago recognized the key distinction between the quantity of labor input and the quantity and quality of output, it has only recently become a major focus of economists' attention (at least partially a consequence, in recent years, of the quality differences in labor output between Japanese and American automobile manufacturers). Conventions about output, forms of organization designed to encourage work participation and cooperation, and attempts to select workers who have an ideological commitment to hard work have all become recent research agendas in the New Industrial Organization. The unique feature of labor markets is that institutions are devised to take into account that the quantity and quality of output are influenced by the attitude of the productive factor – hence morale building is a substitute at the margin for investing in more monitoring.

III

Throughout the discussion above, it was emphasized that the cost of transacting reflects the overall complex of institutions – formal and informal – that make up an economy or, on an even greater scale, a society. This overall structure ultimately shapes the cost of transacting at the individual contract level, and when economists talk about efficient markets, they have simply taken for granted an elaborate framework of constraints. The cost of capital, for example, was in part determined by an elaborate structure of financial intermediaries whose interconnections among consumer credit and mortgage markets, stock markets, and bond markets were constrained by a complex structure of governmental constraints and regulatory agencies, from (in the United States) the Federal Reserve System to state laws and regulatory agencies dealing with everything from branch banking to interest rate ceilings. Moreover, behind the supply of and demand for capital are still other institutions and organizations such as title insurance and credit rating bureaus. Probing still more deeply into the institutional structure reveals political institutions that define formal constraints. The interconnection between the political and economic institutions and organizations is apparent in congressional

committees that are concerned with the capital market and a host of lobbying organizations reflecting interested (and organized) groups in the capital market. The web of interconnections extends to the executive branch as well as to specific regulatory agencies. All of this, of course, has been grist for the new political economy in its attempt to spell out and provide an analytical framework for the institutional structures of Congress and other branches of government.

When we compare the cost of transacting in a Third World country with that in an advanced industrial economy, the costs per exchange in the former are much greater – sometimes no exchange occurs because costs are so high. The institutional structure in the Third World lacks the formal structure (and enforcement) that underpins efficient markets. However, frequently there will exist in Third World countries informal sectors (in effect underground economies) that attempt to provide a structure for exchange. Such structure comes at high costs, however, because the lack of formal property right safeguards restricts activity to personalized exchange systems that can provide self-enforcing types of contracts.[2] But there is more to the issue than simply higher transaction costs in Third World countries. In addition, the institutional framework, which determines the basic structure of production, tends to perpetuate underdevelopment.

Firms come into existence to take advantage of profitable opportunities, which will be defined by the existing set of constraints. With insecure property rights, poorly enforced laws, barriers to entry, and monopolistic restrictions, the profit-maximizing firms will tend to have short time horizons and little fixed capital, and will tend to be small scale. The most profitable businesses may be in trade, redistributive activities, or the black market. Large firms with substantial fixed capital will exist only under the umbrella of government protection with subsidies, tariff protection, and payoffs to the polity – a mixture hardly conducive to productive efficiency.

IV

I conclude this chapter by drawing out some of the implications of the foregoing analysis.

1. The institutional constraints that define the opportunity set of individuals are a complex of formal and informal constraints. They make up an interconnected web that in various combinations shapes choice sets in various contexts. It is easy, given this understanding, to see why institu-

[2]See Hernando de Soto's *The Other Path* (1989) for an insightful analysis of the structure of the Peruvian economy and the characteristics of the large informal sector.

tions are stable and why they typically produce many different margins at which choices are made.

Stability derives from the fact that there are a large number of specific constraints that affect a particular choice, such as those described in the sale of residential property. Significant changes in this institutional framework involve a host of changes in a variety of constraints, not only legal constraints but norms of behavior as well. Although the institutional constraints may not be ideal or efficient for one set of individuals involved in a particular exchange and therefore those parties would like to restructure the institutions, the same set of institutions for other sets of choices may still reflect as efficient a bargain as is possible. Moreover it is the bargaining strength of the individuals and organizations that counts. Hence, only when it is in the interest of those with sufficient bargaining strength to alter the formal rules will there be major changes in the formal institutional framework.

At the same time, the complex of informal and formal constraints makes possible continual incremental changes at particular margins. These small changes in both formal rules and informal constraints will gradually alter the institutional framework over time, so that it evolves into a different set of choices than it began with. Both the stability characteristics of institutions and the marginal incremental changes briefly discussed here will be the subject of Part II.

2. The complex of institutional constraints will result in various mixes of formal and informal constraints, which in turn reflect the costliness of measurement and enforcement. The higher these costs, the more will the exchanging parties invoke informal constraints to shape the exchange, although in the extreme, of course, no exchange will take place at all. Vertical integration offers a partial solution to these costs, always with the caveat that although measurement costs inside organizations will differ from those across markets, they are not necessarily lower. To the extent that informal constraints will dominate forms of exchange, they will typically take the form of devising ways to get around the likelihood of defection by the other party. Therefore self-enforcing contracts, such as those described in Chapter 7, will dominate forms of exchange, although there is the recognition of the limitations that necessarily obtain when third-party enforcement is not possible.

3. Transaction costs are the most observable dimension of the institutional framework that underlies the constraints in exchange. They consist of those costs that go through the market (see Wallis and North, 1986) and therefore are measurable, and of hard-to-measure costs that include time acquiring information, queuing, bribery, and so forth, as well as the losses due to imperfect monitoring and enforcement. These hard-to-measure costs make it difficult to assess precisely the total transaction costs

resulting from a particular institution. Nevertheless, to the degree we are able to do so, we progress in measuring the effectiveness of institutions.

The level of interest rates in capital markets is perhaps the most evident quantitative dimension of the efficiency of the institutional framework, but in Third World countries telephone systems that do not work, the inability to get spare parts, endless production interruptions, long queues and waiting time to get permits, and product variability are overwhelming evidence that an effective institutional infrastructure does not exist.

4. The most important moral to be drawn from this chapter is that the institutional framework plays a major role in the performance of an economy. The sale of the house described earlier in this chapter illustrates what is at stake. A large number of institutions makes possible a mass market in housing and the capital market, the institutions that made for security of property rights, and the large number of voluntary organizations that facilitate the exchange play key roles in the existence of the relatively efficient residential market in the United States as compared to similar housing markets in Third World countries or, in the distant past, in the United States. However, I have been at pains to point out that some institutional constraints raise transaction costs. Therefore, the market overall is a mixed bag of institutions; some increase efficiency and some decrease efficiency. Nevertheless, contrasting the institutional framework in countries such as the United States, England, France, Germany, and Japan with Third World countries or those in the historical past in advanced industrial countries makes clear that this institutional framework is the critical key to the relative success of economies, both cross-sectionally as well as through time. The way institutional constraints evolve through time and thereby determine the performance characteristics of economies is the focus of the rest of this book.